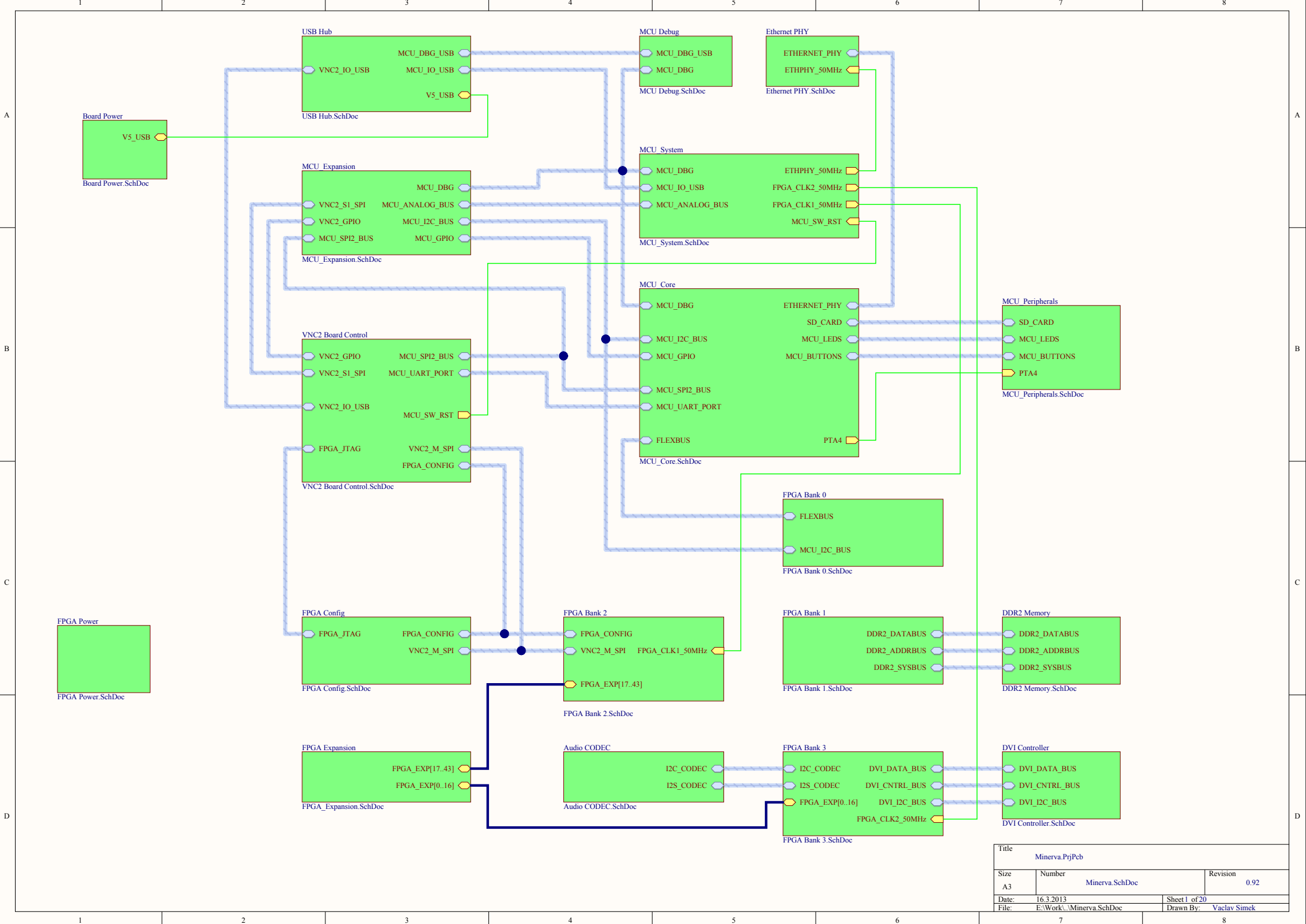
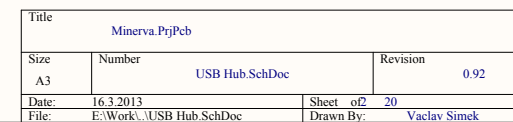
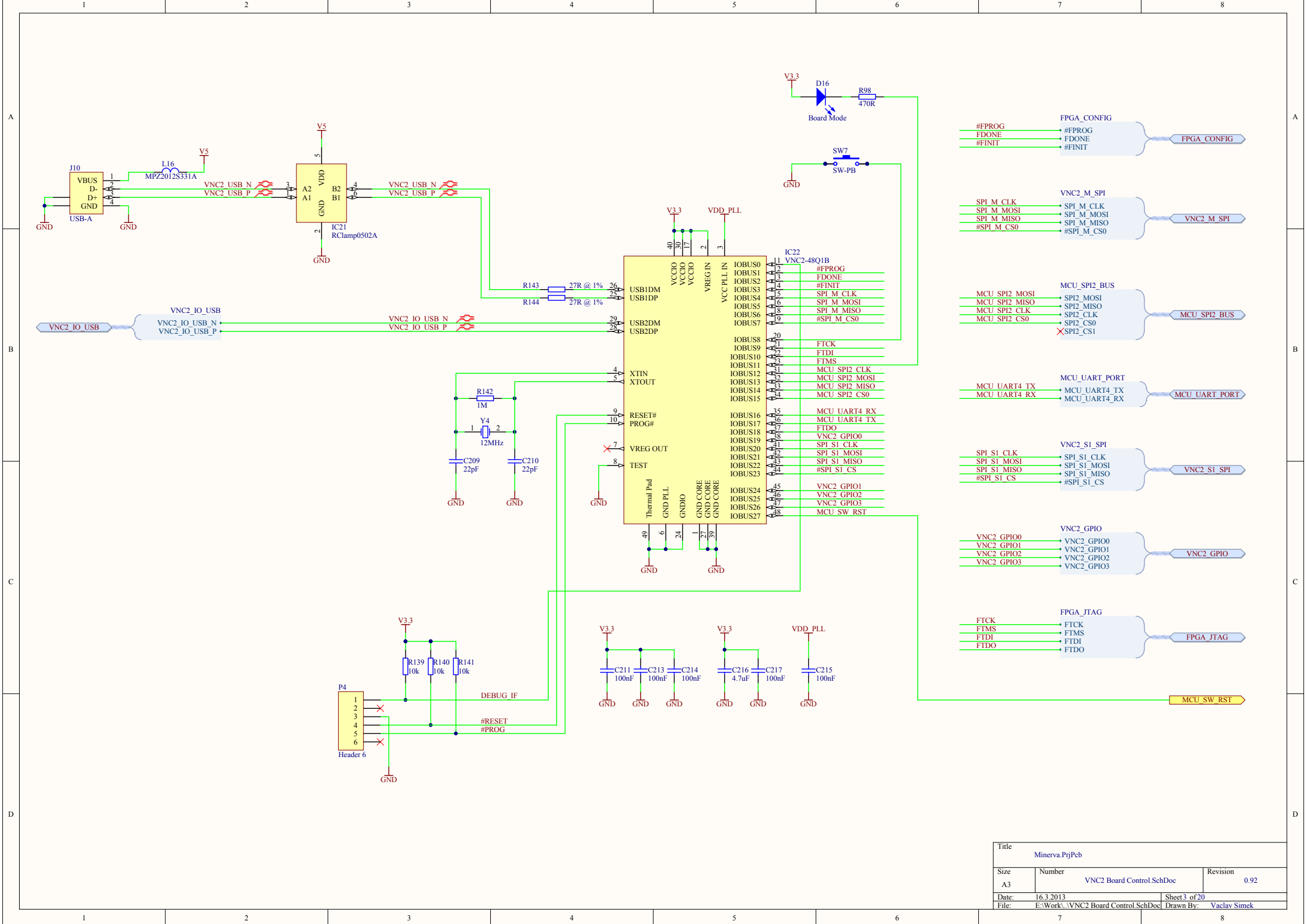


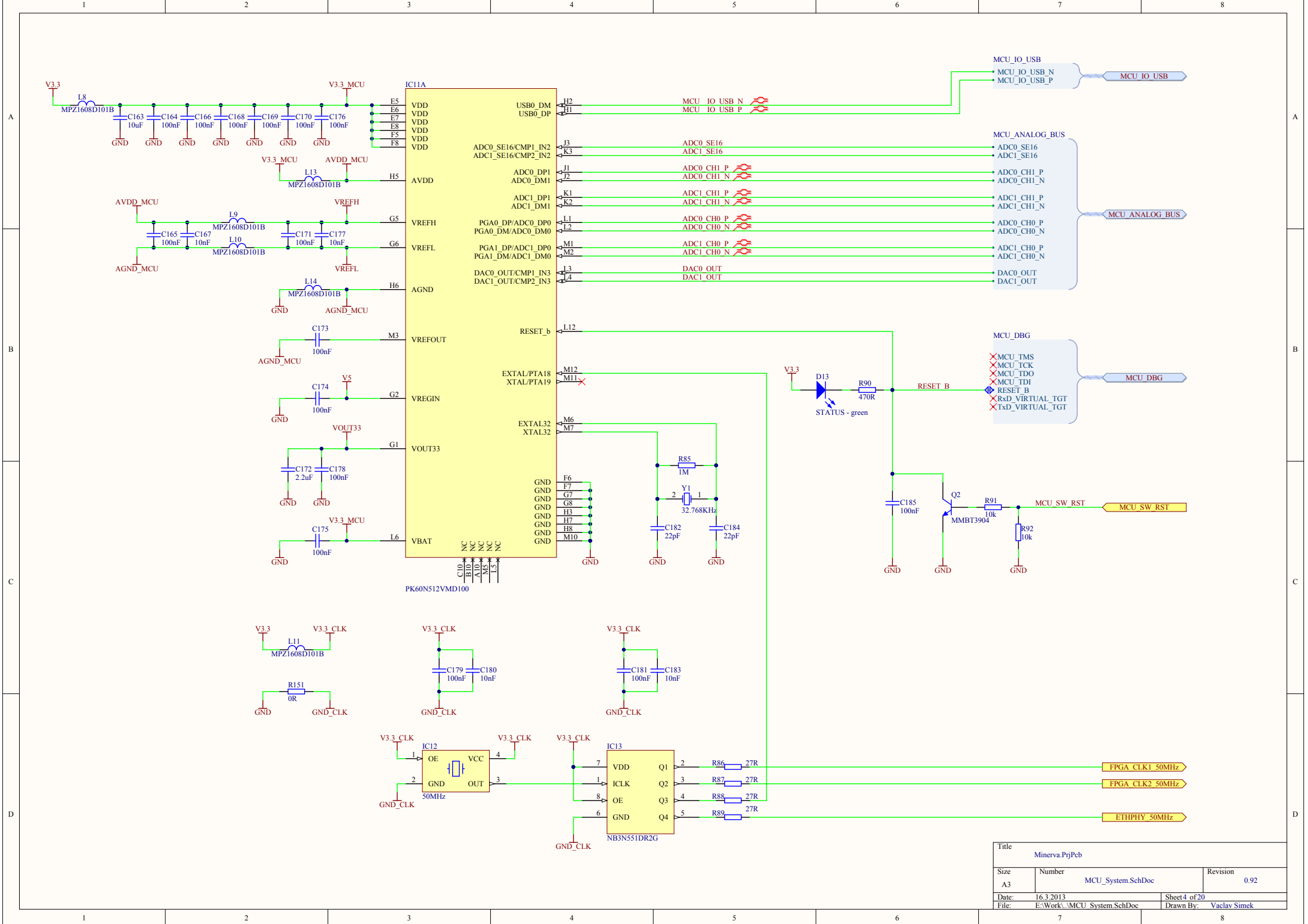
# Schéma obvodového zapojení výukového kitu Minerva



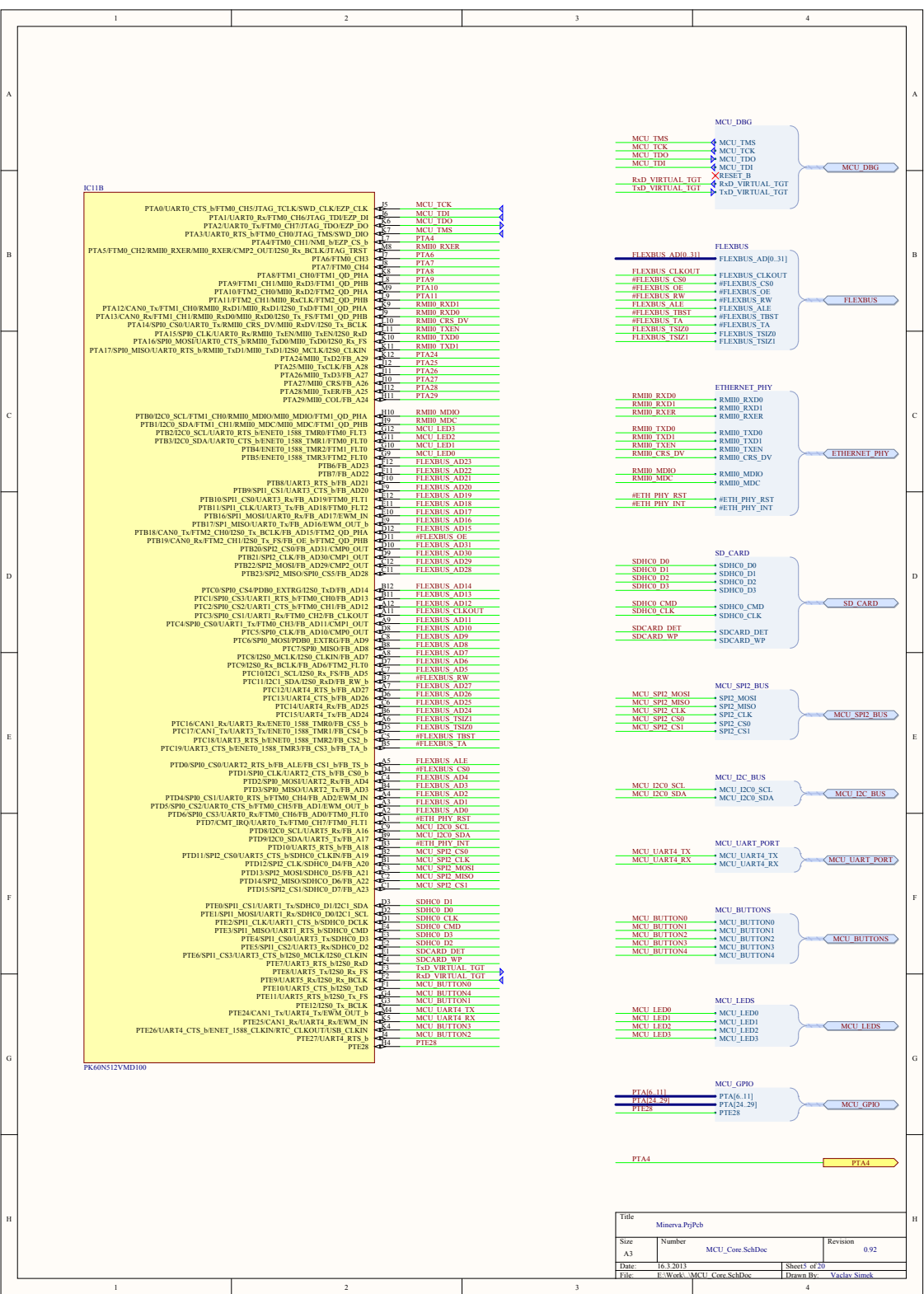
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| Title |                        |                        | Minerva.PtjPcb |
| Size  | Number                 | Revision               |                |
| A3    | Minerva.SchDoc         | 0.92                   |                |
| Date: | 16.3.2013              | Sheet 1 of 20          |                |
| File: | E:\Work\Minerva.SchDoc | Drawn By: Vaclav Simek |                |

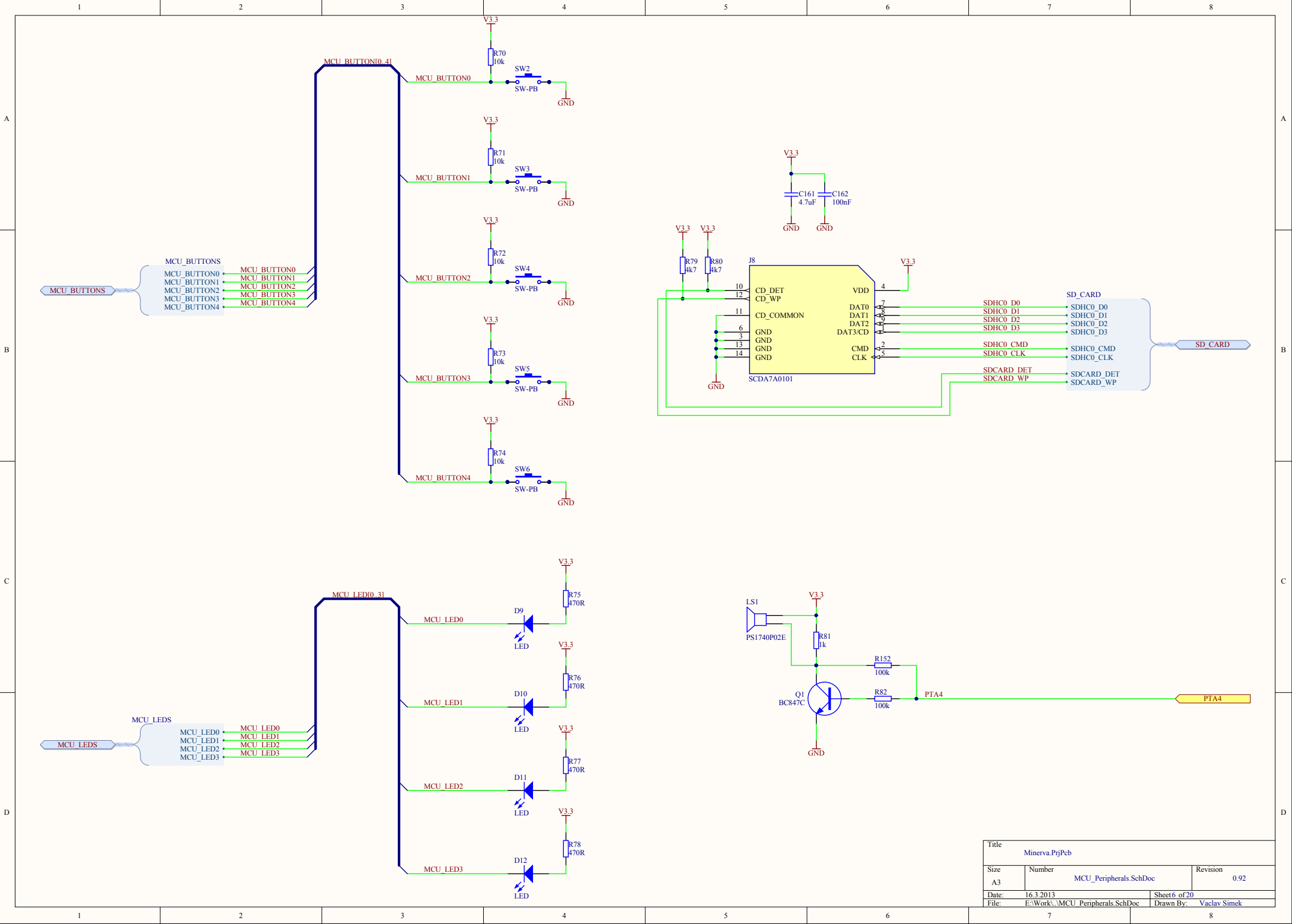






|                |                           |               |              |
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| Title          |                           |               |              |
| Minerva.PtjPcb |                           |               |              |
| Size           | Number                    | Revision      |              |
| A3             | MCU_System.SchDoc         | 0.92          |              |
| Date:          | 16.3.2013                 | Sheet 4 of 20 |              |
| File:          | E:\Work\MCU_System.SchDoc | Drawn By:     | Vaclav Simek |





|                |                                |               |              |
|----------------|--------------------------------|---------------|--------------|
| Title          |                                |               |              |
| Minerva.PtjPcb |                                |               |              |
| Size           | Number                         | Revision      |              |
| A3             | MCU_Peripherals.SchDoc         | 0.92          |              |
| Date:          | 16.3.2013                      | Sheet 6 of 20 |              |
| File:          | E:\Work\MCU_Peripherals.SchDoc | Drawn By:     | Vaclav Simek |

A

B

C

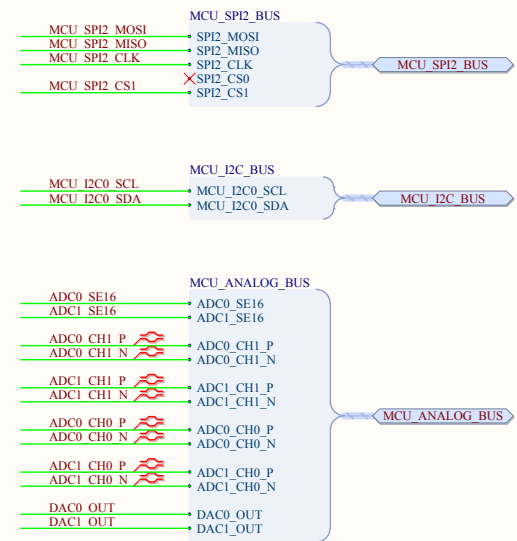
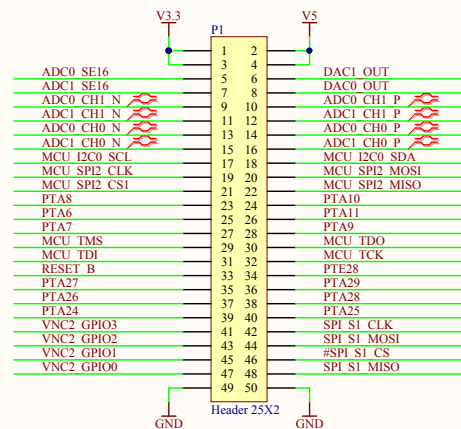
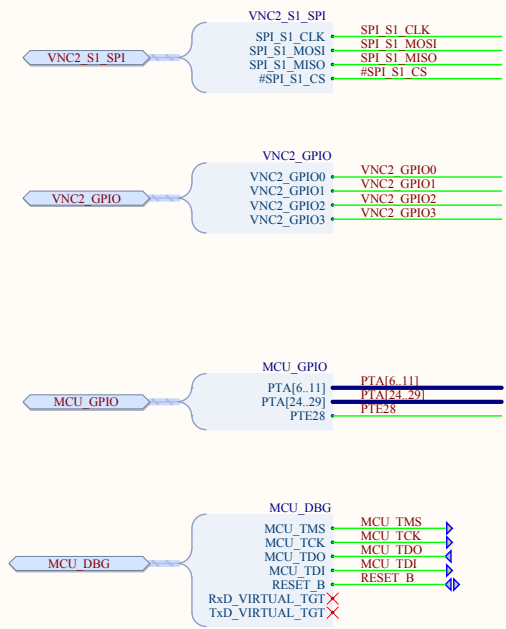
D

A

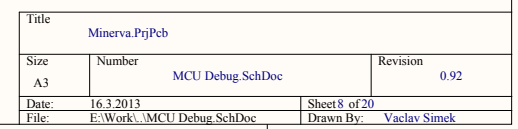
B

C

D



|       |                              |                        |                |
|-------|------------------------------|------------------------|----------------|
| Title |                              |                        | Minerva.PtjPcb |
| Size  | Number                       | Revision               |                |
| A3    | MCU_Expansion.SchDoc         | 0.92                   |                |
| Date: | 16.3.2013                    | Sheet 7 of 20          |                |
| File: | E:\Work\MCU_Expansion.SchDoc | Drawn By: Vaclav Simek |                |



A

A

B

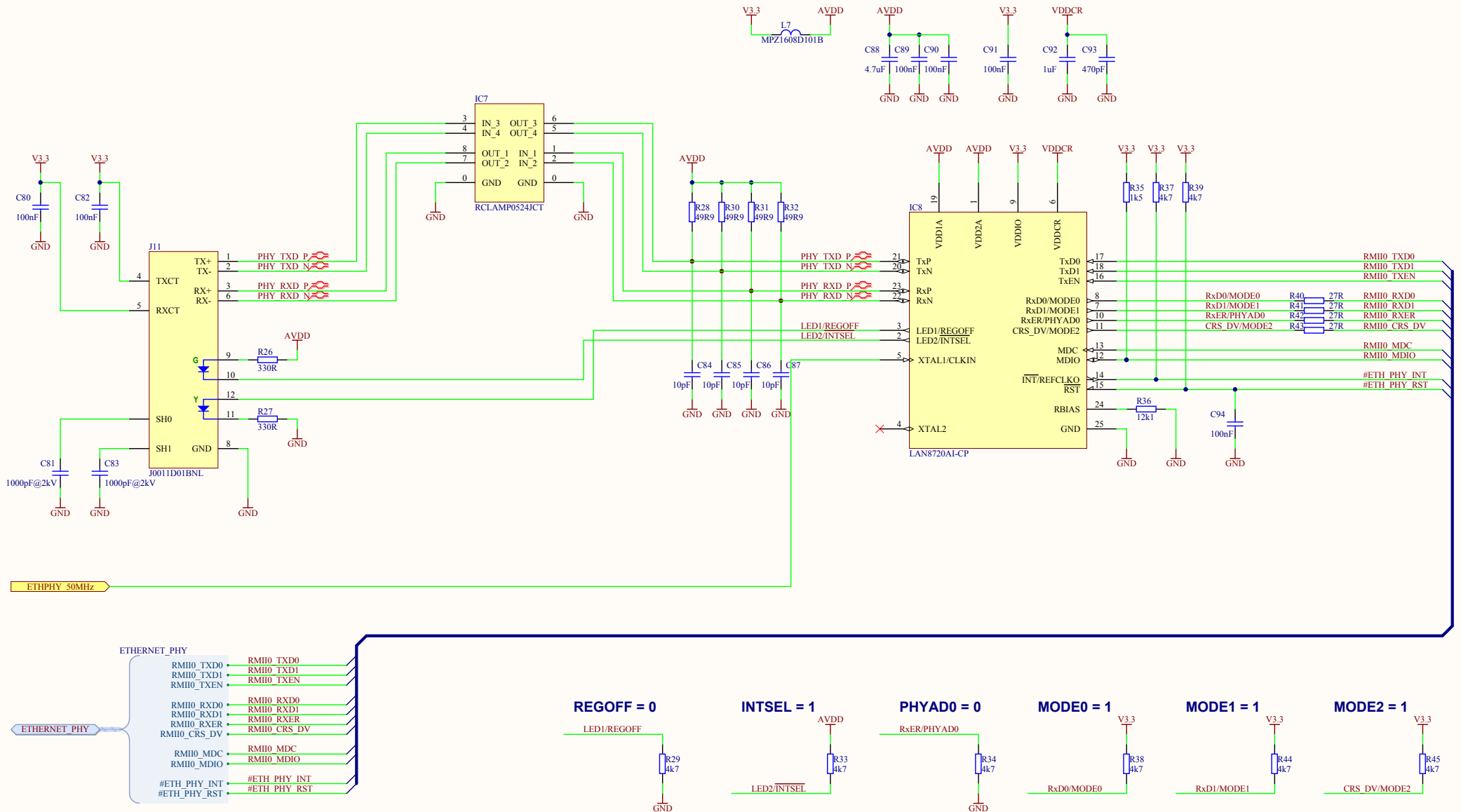
B

C

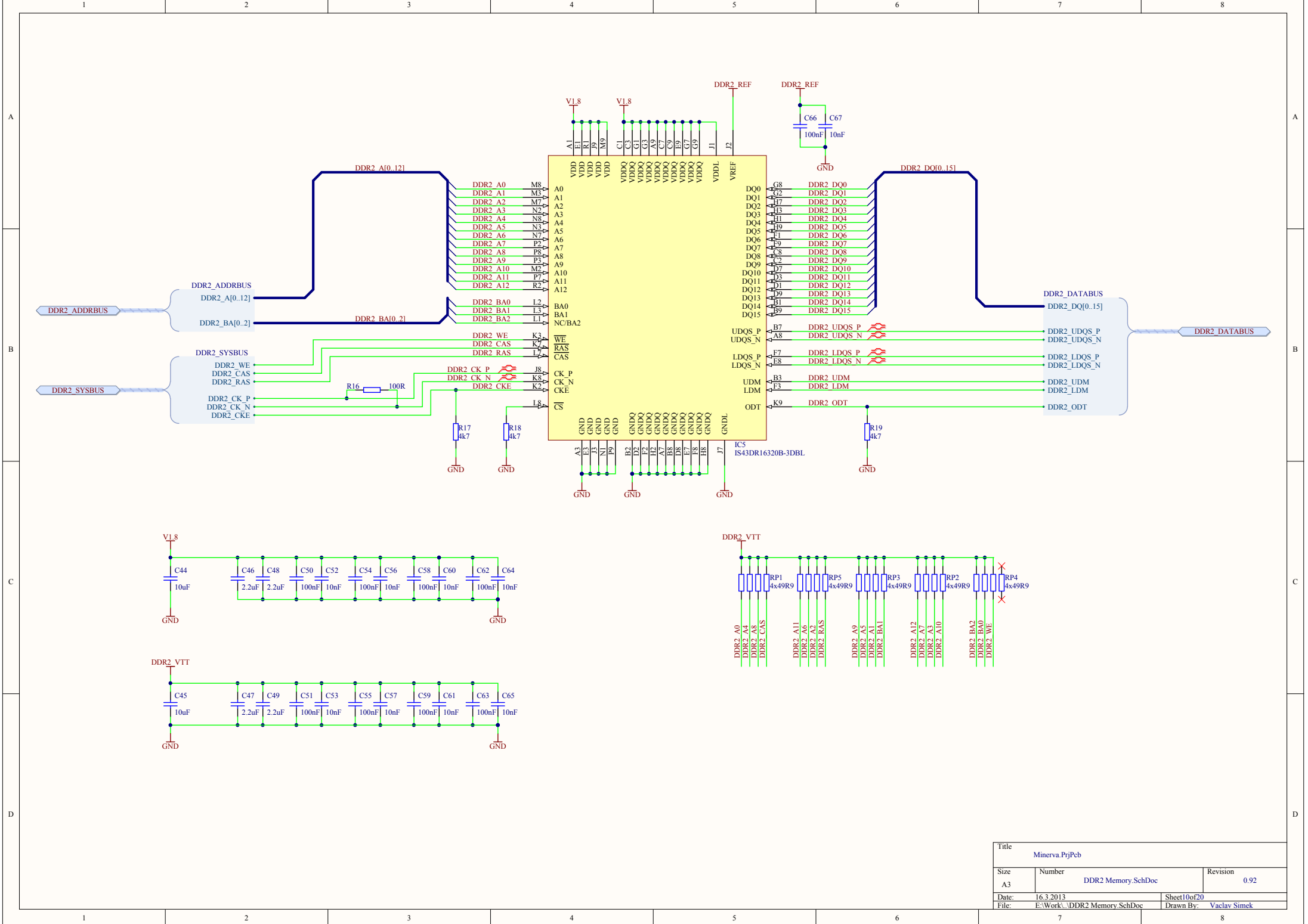
C

D

D



|       |                             |                        |                |
|-------|-----------------------------|------------------------|----------------|
| Title |                             |                        | Minerva.PtjPcb |
| Size  | Number                      | Revision               |                |
| A3    | Ethernet PHY.SchDoc         | 0.92                   |                |
| Date: | 16.3.2013                   | Sheet9 of 20           |                |
| File: | E:\Work\Ethernet PHY.SchDoc | Drawn By: Vaclav Simek |                |



1

2

3

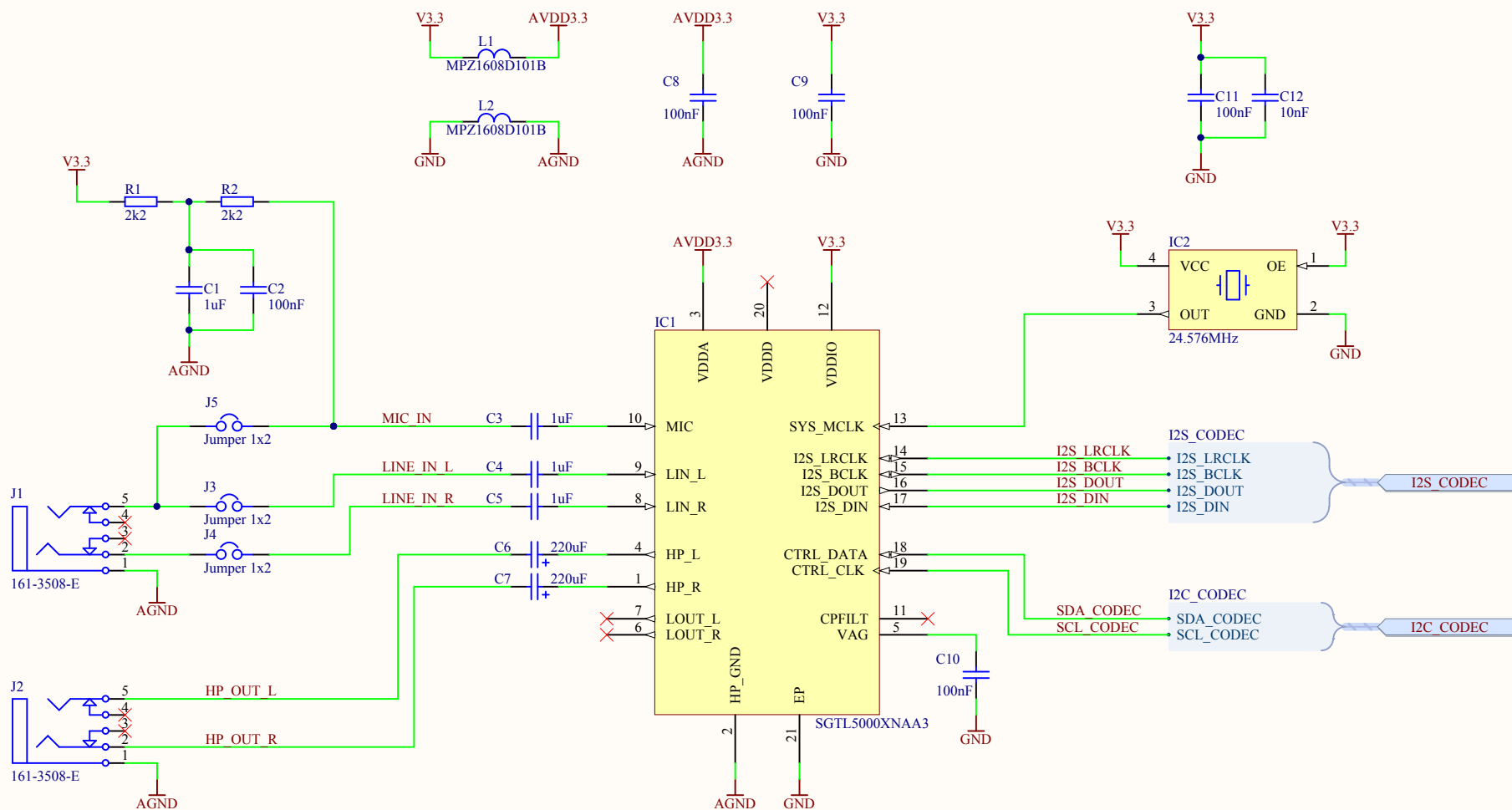
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A

B

C

D



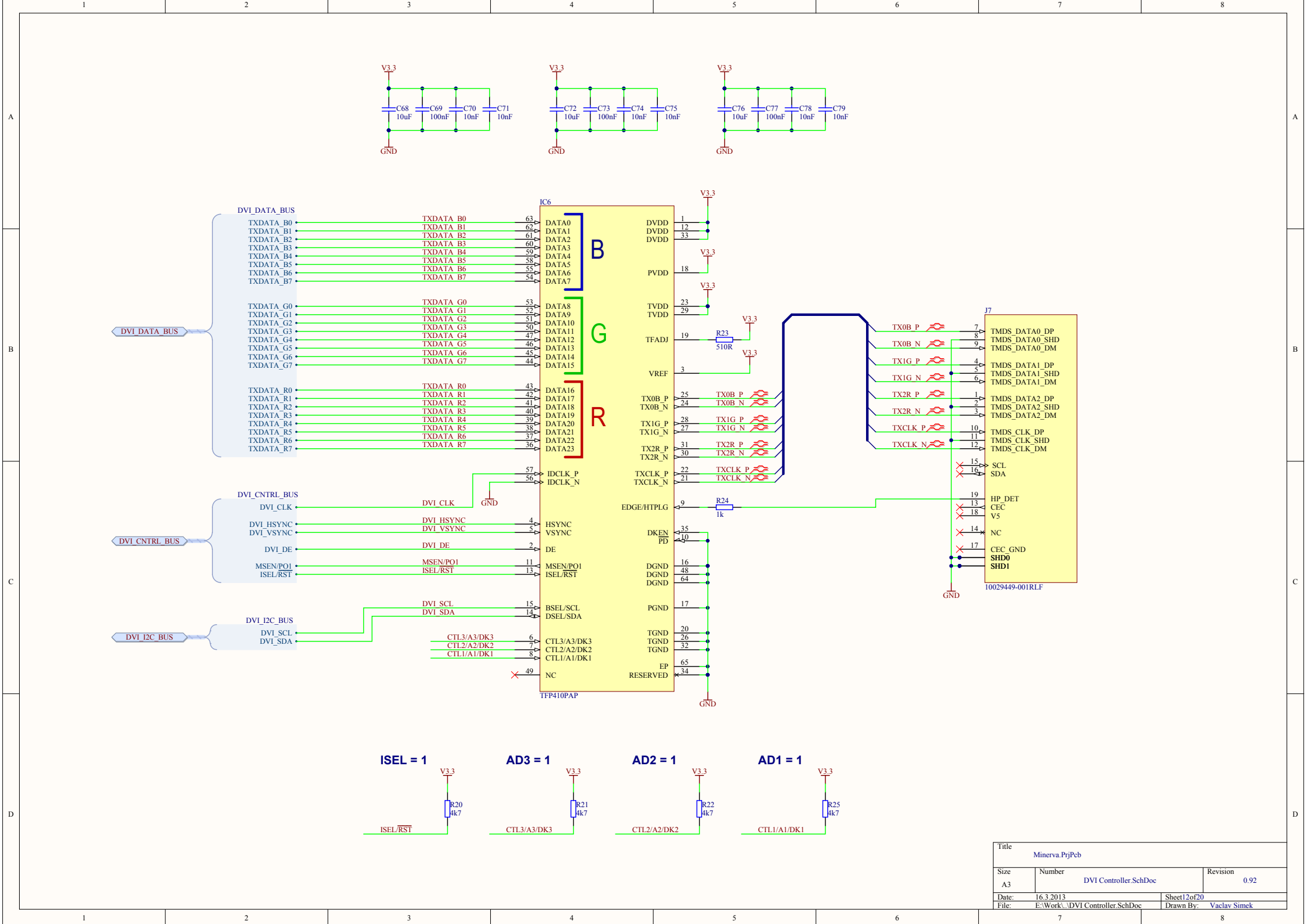
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| Size           | Number                     | Revision               |
| A4             | Audio CODEC.SchDoc         | 0.92                   |
| Date:          | 16.3.2013                  | Sheet 1 of 20          |
| File:          | E:\Work\Audio CODEC.SchDoc | Drawn By: Vaclav Simek |

1

2

3

4



|       |                                   |             |                |
|-------|-----------------------------------|-------------|----------------|
| Title |                                   |             | Minerva.PrjPcb |
| Size  | Number                            | Revision    |                |
| A3    | DVI Controller.SchDoc             | 0.92        |                |
| Date: | 16.3.2013                         | Sheet12of20 |                |
| File: | E:\Work\...\DVI Controller.SchDoc | Drawn By:   | Vaclav Simek   |

IC9A

**BANK 0**

|                  |     |                |
|------------------|-----|----------------|
| IO_L1P_HSWAPEN_0 | D4  | FLEXBUS AD17   |
| IO_L1N_VREF_0    | C4  | FLEXBUS AD22   |
| IO_L2P_0         | A2  | FLEXBUS AD23   |
| IO_L2N_0         | D6  | FLEXBUS AD29   |
| IO_L3P_0         | C6  | FLEXBUS AD30   |
| IO_L3N_0         | B3  | FLEXBUS AD20   |
| IO_L4P_0         | A3  | FLEXBUS AD21   |
| IO_L4N_0         | B4  | FLEXBUS AD18   |
| IO_L5P_0         | A4  | FLEXBUS AD19   |
| IO_L5N_0         | C5  | FLEXBUS AD15   |
| IO_L6P_0         | A5  | FLEXBUS AD16   |
| IO_L6N_0         | B6  | FLEXBUS AD31   |
| IO_L8P_0         | A6  | #FLEXBUS OE    |
| IO_L8N_VREF_0    | C7  | FLEXBUS AD28   |
| IO_L10P_0        | A7  | FLEXBUS AD14   |
| IO_L10N_0        | D8  | MCU I2C0 SDA   |
| IO_L11P_0        | C8  | FLEXBUS AD13   |
| IO_L11N_0        | B8  | FLEXBUS AD12   |
| IO_L33P_0        | A8  | FLEXBUS AD10   |
| IO_L33N_0        | D9  | FLEXBUS CLKOUT |
| IO_L34P_GCLK19_0 | C9  | MCU I2C0_SCL   |
| IO_L34N_GCLK18_0 | B9  | FLEXBUS AD9    |
| IO_L35P_GCLK17_0 | A9  | FLEXBUS AD6    |
| IO_L35N_GCLK16_0 | D11 | FLEXBUS AD27   |
| IO_L36P_GCLK15_0 | C11 | FLEXBUS AD24   |
| IO_L36N_GCLK14_0 | C10 | FLEXBUS AD26   |
| IO_L37P_GCLK13_0 | A10 | FLEXBUS AD5    |
| IO_L37N_GCLK12_0 | G9  | FLEXBUS AD11   |
| IO_L38P_0        | F9  | FLEXBUS AD8    |
| IO_L38N_VREF_0   | B11 | FLEXBUS AD25   |
| IO_L39P_0        | A11 | FLEXBUS TSIZ0  |
| IO_L39N_0        | B12 | FLEXBUS AD7    |
| IO_L41P_0        | A12 | #FLEXBUS RW    |
| IO_L41N_0        | B14 | #FLEXBUS CS0   |
| IO_L62P_0        | A14 | #FLEXBUS TBST  |
| IO_L62N_VREF_0   | F13 | FLEXBUS ALE    |
| IO_L63P_SCP7_0   | E13 | FLEXBUS TSIZ1  |
| IO_L63N_SCP6_0   | C15 | FLEXBUS AD1    |
| IO_L64P_SCP5_0   | A15 | FLEXBUS AD4    |
| IO_L64N_SCP4_0   | D14 | FLEXBUS AD3    |
| IO_L65P_SCP3_0   | C14 | #FLEXBUS TA    |
| IO_L65N_SCP2_0   | B16 | FLEXBUS AD0    |
| IO_L66P_SCP1_0   | A16 | FLEXBUS AD2    |
| IO_L66N_SCP0_0   |     |                |

XC6SLX9-2CSG324C

V3.3

R46

DNP

R47

4k7

GND

FLEXBUS

FLEXBUS\_AD[0..31]

FLEXBUS\_AD[0..31]

FLEXBUS\_CLKOUT

FLEXBUS\_CLKOUT

#FLEXBUS\_CS0

#FLEXBUS\_CS0

#FLEXBUS\_OE

#FLEXBUS\_OE

#FLEXBUS\_RW

#FLEXBUS\_RW

FLEXBUS\_ALE

FLEXBUS\_ALE

#FLEXBUS\_TBST

#FLEXBUS\_TBST

#FLEXBUS\_TA

#FLEXBUS\_TA

FLEXBUS\_TSIZ0

FLEXBUS\_TSIZ0

FLEXBUS\_TSIZ1

FLEXBUS\_TSIZ1

FLEXBUS

MCU\_I2C\_BUS

MCU\_I2C0\_SCL

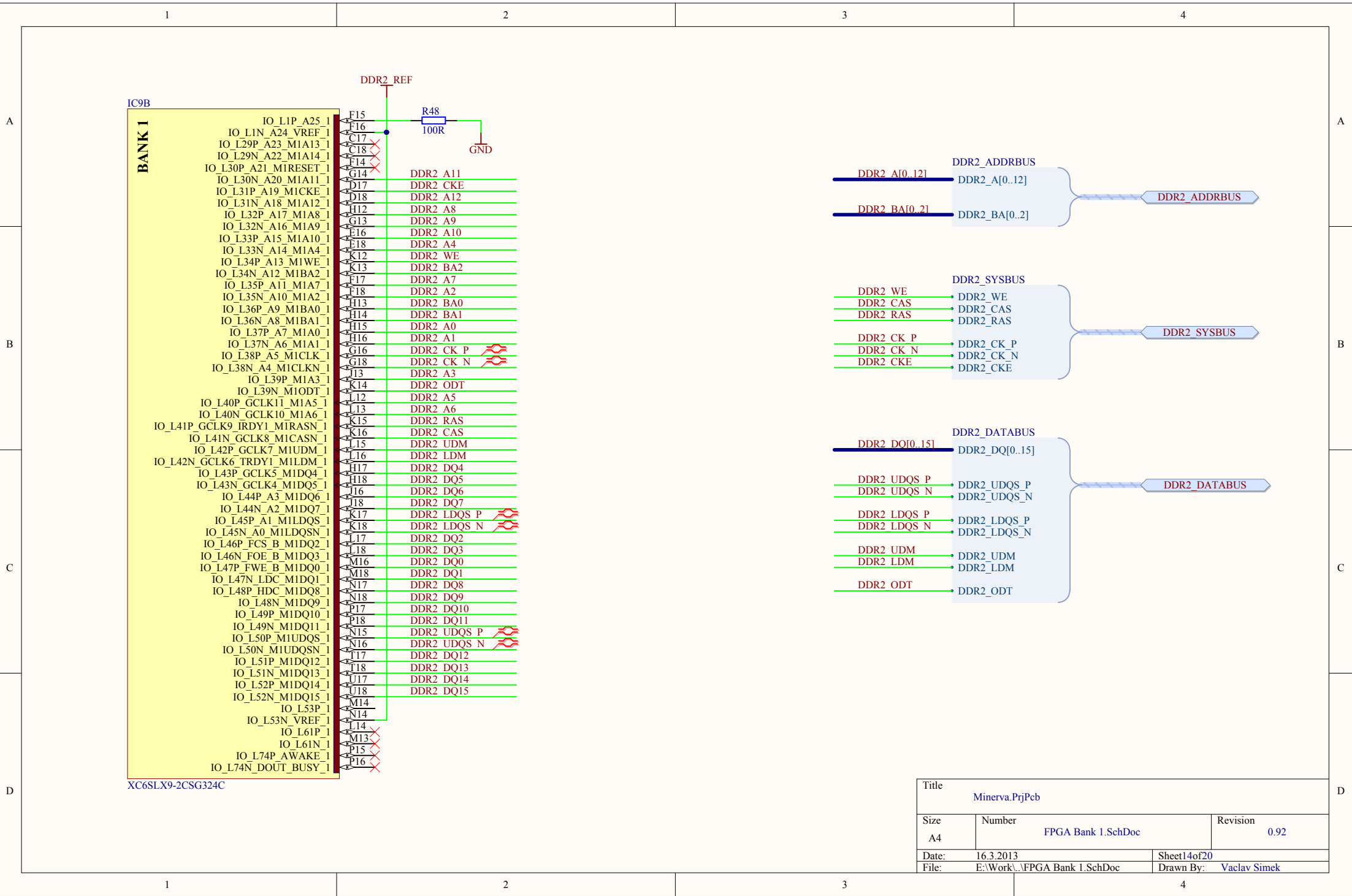
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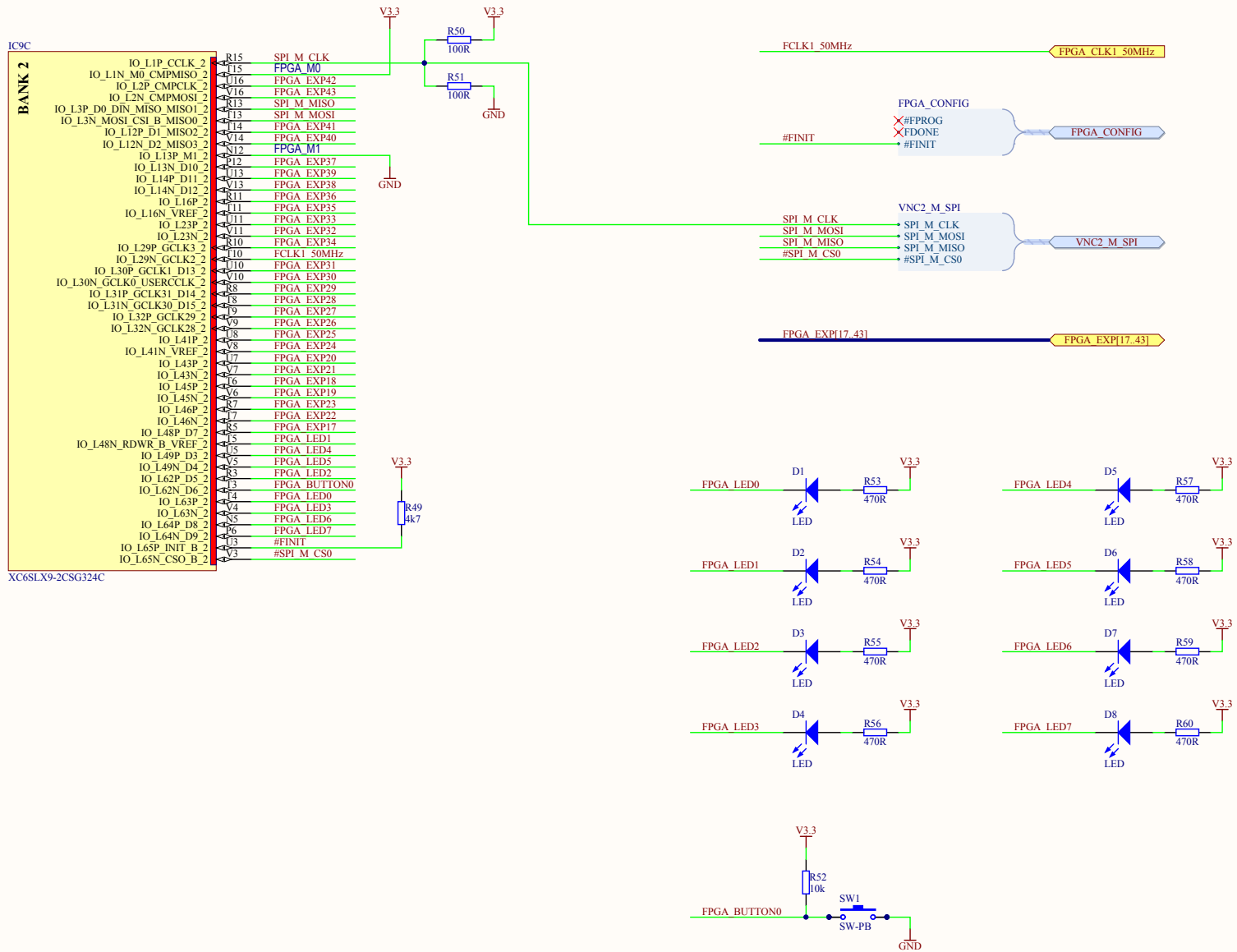
MCU\_I2C0\_SDA

MCU\_I2C0\_SDA

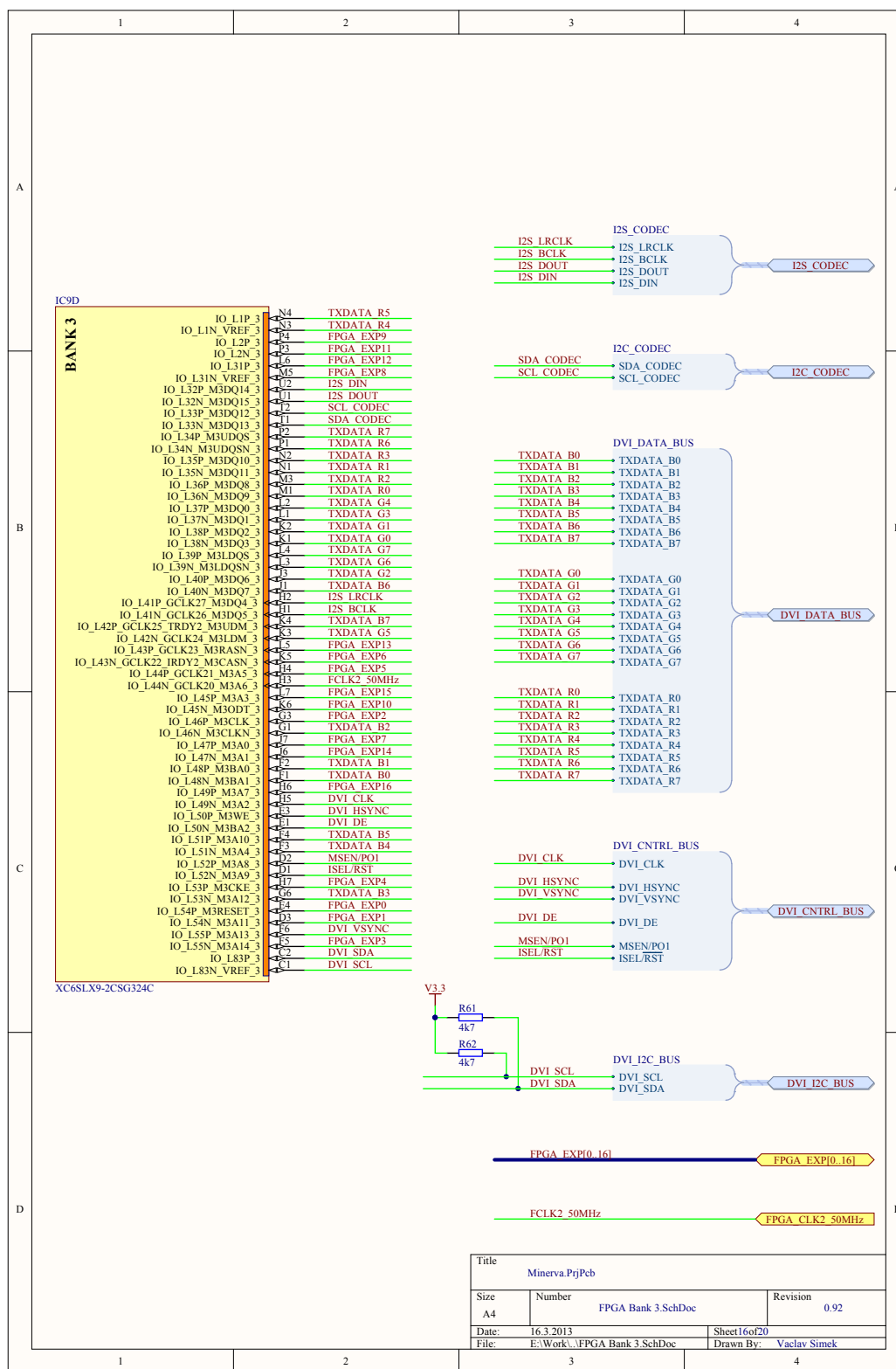
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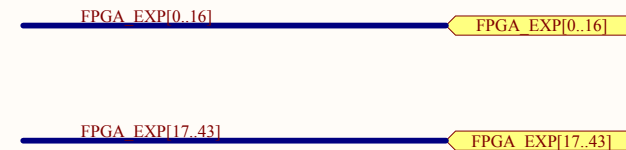
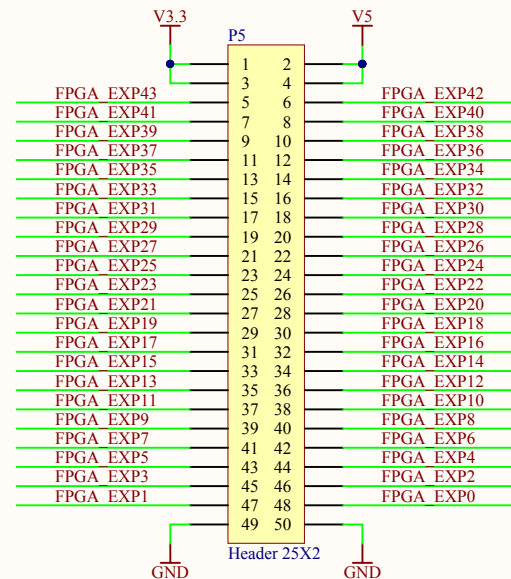
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| Size           | Number                         | Revision               |  |
| A4             | FPGA Bank 0.SchDoc             | 0.92                   |  |
| Date:          | 16.3.2013                      | Sheet13of20            |  |
| File:          | E:\Work\...\FPGA Bank 0.SchDoc | Drawn By: Vaclav Simek |  |



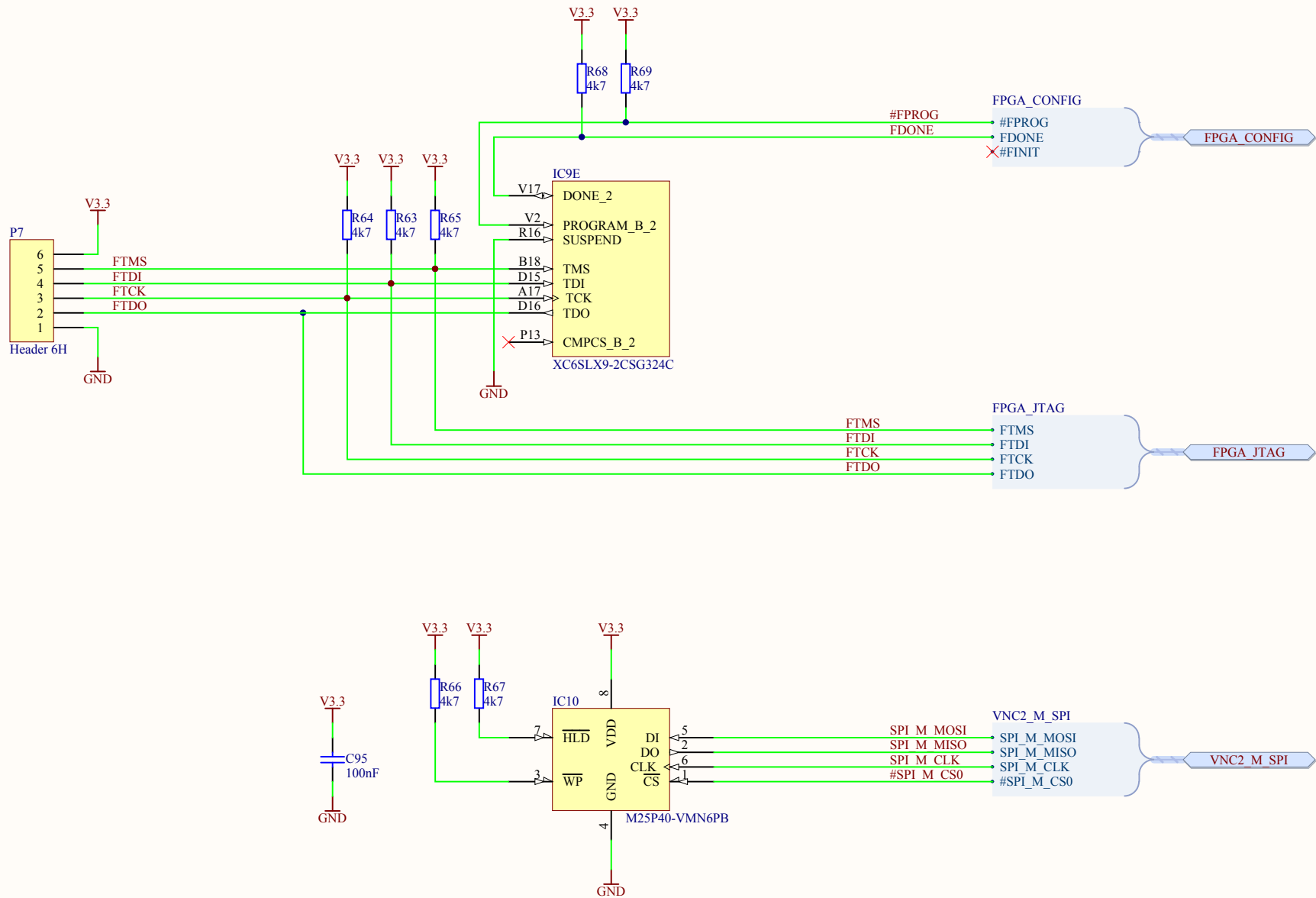


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| Title |                               |                        | Minerva.PrjPcb |
| Size  | Number                        | Revision               |                |
| A3    | FPGA Bank 2.SchDoc            | 0.92                   |                |
| Date: | 16.3.2013                     | Sheet15of20            |                |
| File: | E:\Work\...FPGA Bank 2.SchDoc | Drawn By: Vaclav Simek |                |

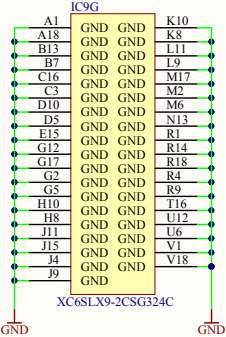
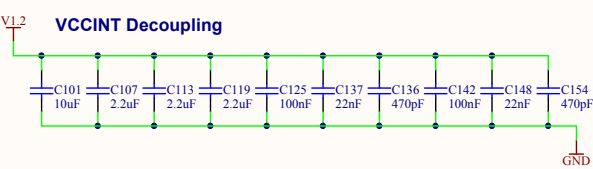
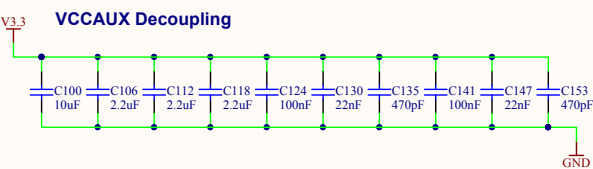
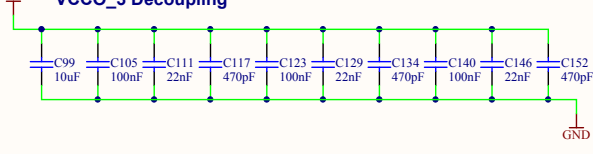
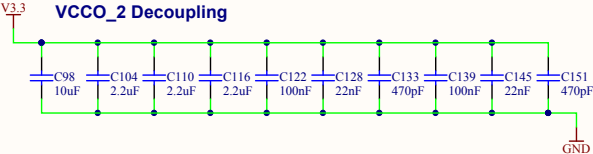
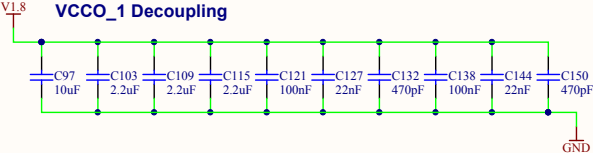
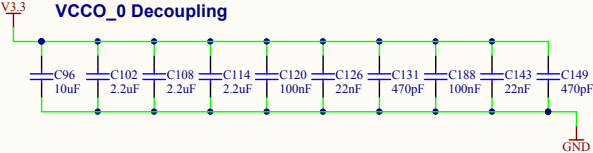
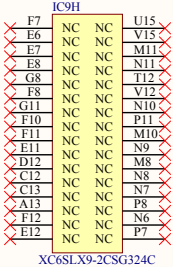
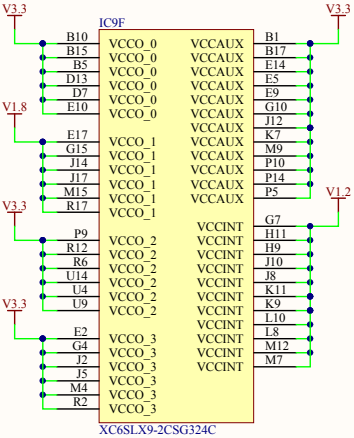




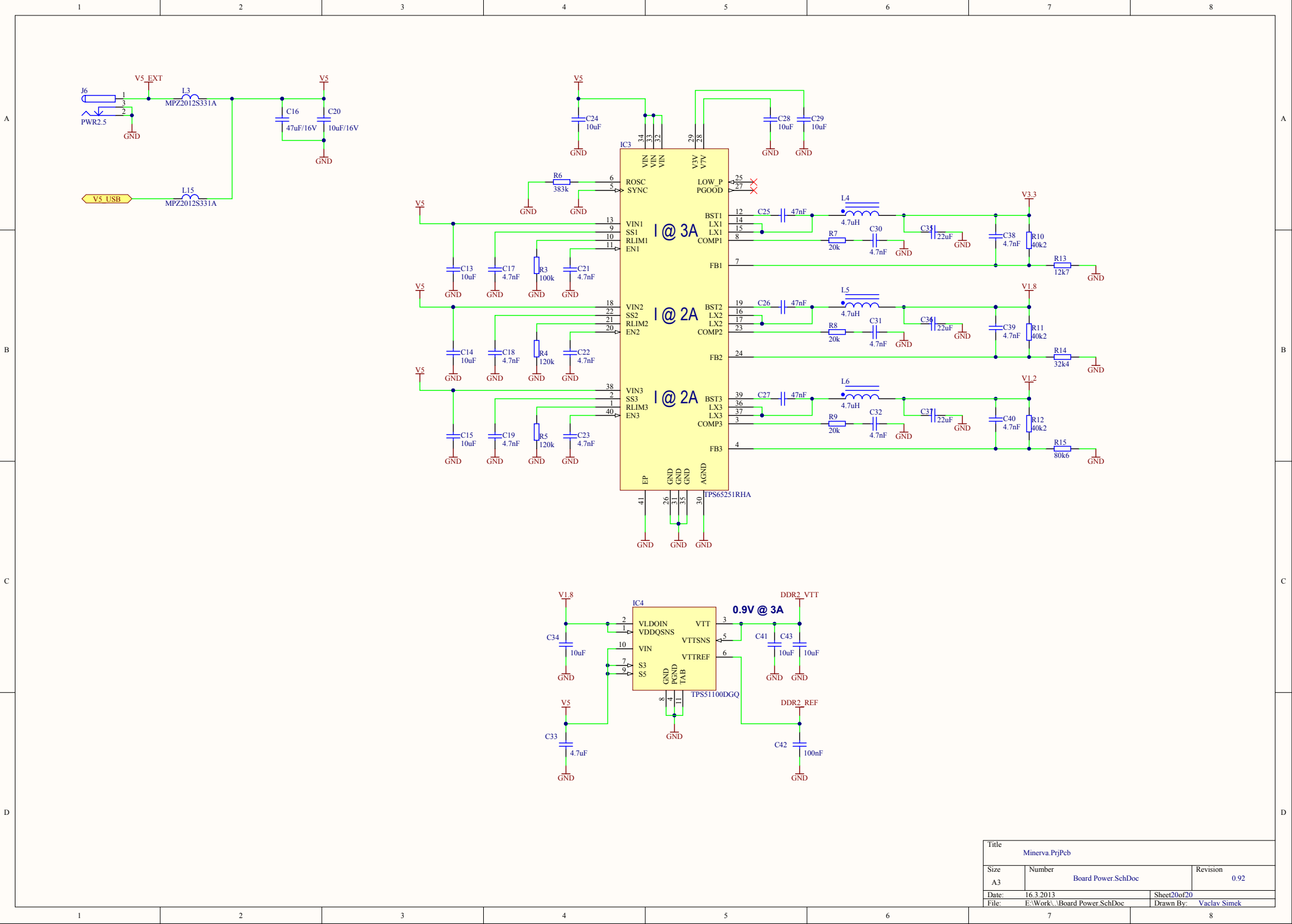
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| Size           | Number                            |  | Revision               |
| A4             | FPGA_Expansion.SchDoc             |  | 0.92                   |
| Date:          | 16.3.2013                         |  | Sheet17of20            |
| File:          | E:\Work\...\FPGA_Expansion.SchDoc |  | Drawn By: Vaclav Simek |



|                |                              |                        |  |
|----------------|------------------------------|------------------------|--|
| Title          |                              |                        |  |
| Minerva.PrjPcb |                              |                        |  |
| Size           | Number                       | Revision               |  |
| A4             | FPGA Config.SchDoc           | 0.92                   |  |
| Date:          | 16.3.2013                    | Sheet18of20            |  |
| File:          | E:\Work\..FPGA Config.SchDoc | Drawn By: Vaclav Simek |  |



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| Title |                              |                        | Minerva.PrjPcb |
| Size  | Number                       | Revision               |                |
| A3    | FPGA Power.SchDoc            | 0.92                   |                |
| Date: | 16.3.2013                    | Sheet19of20            |                |
| File: | E:\Work\...FPGA Power.SchDoc | Drawn By: Vaclav Simek |                |



|                |                            |             |                        |
|----------------|----------------------------|-------------|------------------------|
| Title          |                            |             |                        |
| Minerva.PrjPcb |                            |             |                        |
| Size           | Number                     |             | Revision               |
| A3             | Board Power.SchDoc         |             | 0.92                   |
| Date:          | 16.3.2013                  | Sheet20of20 |                        |
| File:          | E:\Work\Board Power.SchDoc |             | Drawn By: Vaclav Simek |